

ESAT PRACTICE TEST SERIES

ESAT 2024-25 Recall Mock

Mathematics 1 • Physics • Biology

- 27 multiple-choice questions per module
- 40 minutes per module
- Non-calculator
- Hard, recall-informed original practice with full worked solutions

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

ESAT 2024-25 Recall Mock - Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Mathematics 1 answer key

Q1	C	Q2	A	Q3	D	Q4	E	Q5	D	Q6	B	Q7	B	Q8	C	Q9	A
Q10	E	Q11	C	Q12	B	Q13	E	Q14	E	Q15	A	Q16	D	Q17	A	Q18	B
Q19	D	Q20	A	Q21	D	Q22	C	Q23	E	Q24	B	Q25	D	Q26	E	Q27	C

Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

M1-01

Ratio and replacement

Answer **C**

Question 1

A tank contains an alloy in which copper and zinc are in the mass ratio **5 : 3**. Exactly **8 kg** of the well-mixed alloy is removed and replaced by **8 kg** of pure zinc. The new copper-to-zinc ratio is **3 : 2**.

What was the original mass of alloy in the tank?

A 80 kg

B 192 kg

C 200 kgD $\frac{200}{9}$ kg

E 320 kg

WORKED SOLUTION

Let the original mass be M kg. Removing 8 kg of the well-mixed alloy leaves

$$\text{copper} = \frac{5}{8}(M - 8), \quad \text{zinc} = \frac{3}{8}(M - 8).$$

After 8 kg of zinc is added,

$$\frac{\frac{5}{8}(M - 8)}{\frac{3}{8}(M - 8) + 8} = \frac{3}{2}.$$

Hence $10(M - 8) = 9(M - 8) + 192$, so $M = 200$.

Therefore the correct option is **C**.

Trap check: Option B ignores the removed alloy before the zinc is added; option A treats the removal as changing the copper-to-zinc ratio even though a well-mixed sample preserves it; option D reverses the final ratio; option E treats the removed mass as pure copper.

Question 2

For positive integers a and b , the highest common factor of a and b is 6 , and their lowest common multiple is 180 .

How many *ordered* pairs (a, b) satisfy these conditions?

A 8**B** 4**C** 6**D** 3**E** 16**WORKED SOLUTION**

Write $a = 6x$ and $b = 6y$, where x and y are coprime. Then

$$6xy = 180 \implies xy = 30 = 2 \cdot 3 \cdot 5.$$

Each prime factor must be assigned wholly to either x or y . There are $2^3 = 8$ assignments, and order matters.

Therefore the correct option is **A**.

Trap check: Option B counts unordered pairs, C omits the assignments with one reduced factor equal to 1, D merely counts the distinct primes, and E incorrectly includes negative ordered pairs despite the positivity condition.

Question 3

A vehicle travels a route of length D . On one day it covers the first third of the route at speed v and the remainder at speed $2v$, taking **48** minutes in total.

On another day it covers the first half at speed $4v/3$, stops for **6** minutes, and then covers the second half at constant speed kv . The total time is again **48** minutes.

What is k ?

A 2

B $\frac{9}{4}$

C $\frac{5}{2}$

D $\frac{12}{5}$

E $\frac{8}{3}$

WORKED SOLUTION

The first journey gives

$$\frac{D/3}{v} + \frac{2D/3}{2v} = \frac{2D}{3v} = 48,$$

so $D/v = 72$ minutes. On the second day the first half takes

$$\frac{D/2}{4v/3} = \frac{3}{8} \frac{D}{v} = 27 \text{ minutes.}$$

After the six-minute stop, **15** minutes remain. Therefore

$$\frac{D/2}{kv} = \frac{36}{k} = 15, \quad k = \frac{12}{5}.$$

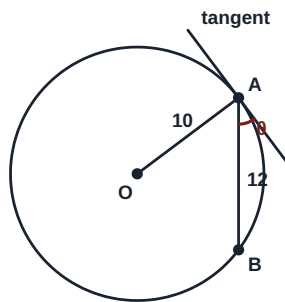
Therefore the correct option is **D**.

Trap check: Do not average segment speeds without checking the associated times; here the arithmetic mean happens to agree because the two segment times are equal.

Question 4

A circle has centre O , radius 10 , and chord AB of length 12 . A tangent is drawn at A . Let θ be the acute angle between the tangent and chord AB .

What is $\tan \theta$?



A $\frac{4}{3}$

B $\frac{3}{5}$

C $\frac{4}{5}$

D $\frac{5}{4}$

E $\frac{3}{4}$

WORKED SOLUTION

Let M be the midpoint of AB . The perpendicular from the centre to a chord bisects it, so $AM = 6$. Hence

$$OM = \sqrt{10^2 - 6^2} = 8.$$

If $\alpha = \angle OAB$, then $\tan \alpha = 8/6 = 4/3$. Since the tangent is perpendicular to OA , $\theta = 90^\circ - \alpha$, and therefore

$$\tan \theta = \frac{1}{\tan \alpha} = \frac{3}{4}.$$

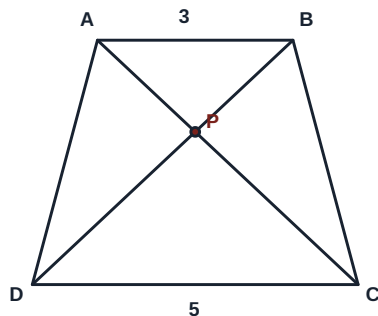
Therefore the correct option is E.

Trap check: The value $4/3$ is $\tan \angle OAB$, not the tangent–chord angle requested.

Question 5

In trapezium $ABCD$, $AB \parallel CD$. The diagonals AC and BD meet at P , and $AB : CD = 3 : 5$. The area of triangle APB is 27.

What is the area of triangle CPD ?



A 45

B 125

C $\frac{243}{25}$

D 75

E 192

WORKED SOLUTION

Triangles APB and CPD are similar: they have equal vertically opposite angles at P , and corresponding angles from $AB \parallel CD$. Their linear scale factor is $3 : 5$, so their areas are in the ratio $9 : 25$.

$$27 : \text{area}(CPD) = 9 : 25.$$

Therefore $\text{area}(CPD) = 27 \times 25/9 = 75$.

Therefore the correct option is **D**.

Trap check: Option A uses the linear scale factor, B cubes it, C reverses the area factor, and E gives the area of the whole trapezium rather than triangle CPD .

Question 6

The sequence begins

$$7, 14, 25, 40, \dots$$

and has constant second differences. Let n be the smallest index for which the n th term exceeds 500, and let e be the amount by which that term exceeds 500.

Which pair (n, e) is correct?

A (15, 31)

B (16, 32)

C (16, 31)

D (15, 32)

E (17, 69)

WORKED SOLUTION

The first differences are 7, 11, 15, ..., so the second difference is 4. Fitting $u_n = 2n^2 + bn + c$ gives

$$u_n = 2n^2 + n + 4.$$

Now $u_{15} = 469$, whereas $u_{16} = 532$. Thus $n = 16$, and the excess is $e = 532 - 500 = 32$.

Therefore the correct option is B.

Trap check: A and D stop one term too early; C subtracts incorrectly after finding the correct index; E moves to the next term after the threshold has already been crossed.

Question 7

A sample has recorded mass **2.400 kg**, correct to the nearest **0.010 kg**, and recorded volume **800 cm³**, correct to the nearest **10 cm³**.

If its density is $\rho \text{ g cm}^{-3}$, which statement gives the exact bounds for ρ ?

A $\frac{479}{159} \leq \rho < \frac{481}{161}$

B $\frac{479}{161} < \rho < \frac{481}{159}$

C $\frac{480}{161} \leq \rho < \frac{480}{159}$

D $\frac{2395}{800} \leq \rho < \frac{2405}{800}$

E $2.995 \leq \rho < 3.005$

WORKED SOLUTION

In grams and cubic centimetres,

$$2395 \leq m < 2405, \quad 795 \leq V < 805.$$

For positive quantities, m/V is smallest as m takes its lower bound and V approaches its excluded upper bound; it is greatest as m approaches its excluded upper bound and V takes its lower bound. Hence

$$\frac{2395}{805} < \rho < \frac{2405}{795}.$$

Dividing numerator and denominator by 5 gives

$$\frac{479}{161} < \rho < \frac{481}{159}.$$

Therefore the correct option is B.

Trap check: The lower endpoint is also excluded: the recorded-volume interval has $V < 805$, so density can approach $2395/805$ but cannot equal it.

Question 8

A bag contains **4** red, **3** blue and **2** green counters. Two counters are drawn at random without replacement.

Given that the counters are different colours, what is the probability that one of them is green?

A $\frac{5}{13}$

B $\frac{6}{13}$

C $\frac{7}{13}$

D $\frac{7}{18}$

E $\frac{14}{27}$

WORKED SOLUTION

Count unordered pairs. The number of different-colour pairs is

$$4 \cdot 3 + 4 \cdot 2 + 3 \cdot 2 = 26.$$

Of these, the pairs involving green number $4 \cdot 2 + 3 \cdot 2 = 14$. Hence the conditional probability is

$$\frac{14}{26} = \frac{7}{13}.$$

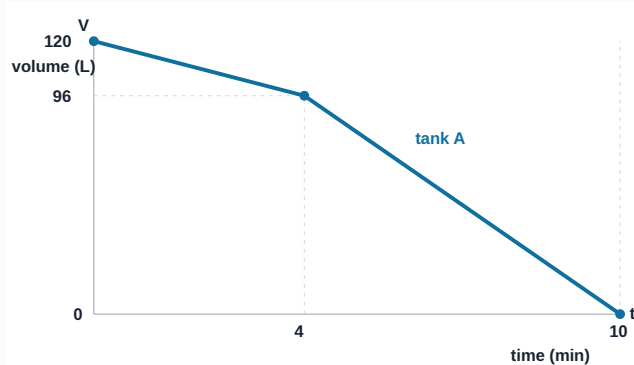
Therefore the correct option is C.

Trap check: The denominator must contain only different-colour pairs, not all $\binom{9}{2}$ pairs.

Question 9

The graph shows the volume V litres of water remaining in tank A, t minutes after draining begins. At $t = 0$, an empty tank B starts filling at a constant rate of **12** litres per minute.

At what time do the two tanks contain the same volume?

**A** $\frac{40}{7}$ min**B** 5 min**C** 6 min**D** $\frac{20}{3}$ min**E** 8 min**WORKED SOLUTION**

At $t = 4$, tank A contains **96** L. From $t = 4$ to $t = 10$, its volume falls by **96** L in **6** min, so the rate is **16** L/min. On this segment,

$$V_A = 96 - 16(t - 4) = 160 - 16t.$$

Tank B contains $V_B = 12t$. Equating volumes gives

$$160 - 16t = 12t \implies t = \frac{40}{7}.$$

This lies between **4** and **10**, so the correct segment was used.

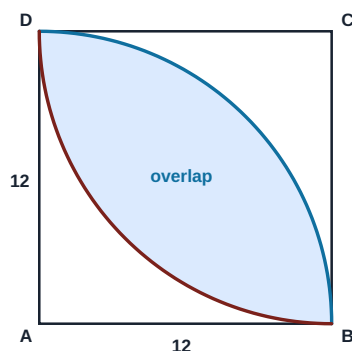
Therefore the correct option is A.

Trap check: The first segment's rule gives a time outside its interval; the change of gradient at 4 minutes matters.

Question 10

A square $ABCD$ has side length **12**. Inside it, quarter-circles of radius **12** are drawn with centres **A** and **C**. The shaded lens is the region lying inside both quarter-circles.

What is the exact area of the shaded region?



A $36\pi - 72$

B 72π

C $144 - 36\pi$

D $72\pi - 72$

E $72\pi - 144$

WORKED SOLUTION

Each arc BD subtends 90° at its centre because adjacent sides of the square are perpendicular. One half of the lens is a 90° sector of radius **12**, less a right triangle with legs **12**:

$$\frac{1}{4}\pi(12)^2 - \frac{1}{2}(12)(12) = 36\pi - 72.$$

The lens has two congruent halves, so its area is

$$2(36\pi - 72) = 72\pi - 144.$$

Therefore the correct option is **E**.

Trap check: A uses only half the lens, B subtracts no triangles, C takes an incorrect square complement, and D subtracts only one of the two right triangles.

Question 11

Let $u = \sqrt{3} + \sqrt{2}$ and $v = \sqrt{3} - \sqrt{2}$.

What is the value of

$$\frac{u^6 + v^6}{u^2 + v^2} ?$$

A 10

B 94

C 97

D 100

E 103

F 970

WORKED SOLUTION

Since $uv = 1$ and $u + v = 2\sqrt{3}$,

$$u^2 + v^2 = (u + v)^2 - 2uv = 10.$$

Using $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$ with $a = u^2$, $b = v^2$,

$$u^6 + v^6 = 10^3 - 3(uv)^2(10) = 1000 - 30 = 970.$$

The required quotient is $970/10 = 97$.

Therefore the correct option is **C**.

Trap check: Option A stops at $u^2 + v^2$, B incorrectly uses $(uv)^2 = 2$, D omits the cubic cross-term, and E changes its sign.

Question 12

The point P lies on the line $y = 2x + 1$ and is equidistant from $A = (0, 4)$ and $B = (6, 0)$. The origin is O .

What is OP^2 ?

A 169

B 218

C 196

D 224

E 49

WORKED SOLUTION

Write $P = (x, 2x + 1)$. Equating squared distances avoids square roots:

$$x^2 + (2x - 3)^2 = (x - 6)^2 + (2x + 1)^2.$$

The quadratic terms cancel, leaving $-12x + 9 = -8x + 37$. Hence $-4x = 28$, so $x = -7$. Therefore $P = (-7, -13)$, and

$$OP^2 = (-7)^2 + (-13)^2 = 49 + 169 = 218.$$

Therefore the correct option is B.

Trap check: A uses only the square of the y -coordinate, C omits x^2 , D introduces a cross-term as if calculating $(x + y)^2$, and E reports only x^2 .

Question 13

Two distinct cards are selected at random from cards numbered **1** to **8**. It is known that their product is even and their sum is **not** divisible by **3**.

What is the probability that their product is divisible by **4**?

A $\frac{9}{22}$

B $\frac{1}{2}$

C $\frac{5}{14}$

D $\frac{7}{11}$

E $\frac{9}{14}$

WORKED SOLUTION

Use unordered pairs. There are **22** pairs with an even product. Of the **10** pairs whose sum is divisible by **3**, two have odd product, so eight are excluded from those **22**. The conditioned sample space therefore has $22 - 8 = 14$ pairs.

The nine admissible pairs with product divisible by **4** are

$$(2, 6), (2, 8), (1, 4), (3, 4), (4, 6), (4, 7), (6, 8), (3, 8), (5, 8).$$

The required probability is **9/14**.

Therefore the correct option is E.

Trap check: A uses the 22 even-product pairs as the denominator and ignores the sum condition; B assumes an unsupported half-and-half split; C counts only a restricted subset centred on the cards 4 and 6; D counts products divisible by 4 before enforcing the sum condition.

Question 14

Let

$$N = 2^4 \cdot 3^3 \cdot 5^2.$$

How many positive divisors of N are divisible by exactly one of **12** and **18**?**A** 9**B** 6**C** 18**D** 33**E** 15**F** 51**WORKED SOLUTION**Divisors divisible by $12 = 2^2 \cdot 3$ number

$$(3)(3)(3) = 27.$$

Divisors divisible by $18 = 2 \cdot 3^2$ number

$$(4)(2)(3) = 24.$$

Those divisible by both are divisible by $\text{lcm}(12, 18) = 36 = 2^2 3^2$, giving

$$(3)(2)(3) = 18.$$

To count divisors in exactly one set, subtract the intersection twice:

$$27 + 24 - 2(18) = 15.$$

Therefore the correct option is E.**Trap check:** Options A and B count divisors in only one named set separately, C counts the intersection, and D is the ordinary inclusion–exclusion total for at least one condition.

Question 15

Containers **A**, **B**, **C** initially hold liquid in the ratio **1 : 3 : 5**. Half of the liquid in **A** is transferred to **B**. Then half of the resulting amount in **B** is transferred to **C**. Finally, **C** contains **100** litres more than **A**.

How much liquid was there originally in total?

A 144 L**B** 200 L**C** 150 L**D** 180 L**E** 108 L**WORKED SOLUTION**

Let the initial amounts be **k**, **3k**, **5k**. After the first transfer,

$$A = \frac{1}{2}k, \quad B = \frac{7}{2}k.$$

Half of this new amount in **B** is $\frac{7k}{4}$, so

$$B = \frac{7}{4}k, \quad C = 5k + \frac{7}{4}k = \frac{27}{4}k.$$

Thus $C - A = (27/4 - 1/2)k = 25k/4 = 100$, giving $k = 16$. The original total was $9k = 144$ L.

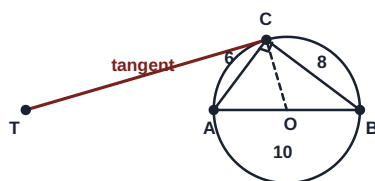
Therefore the correct option is **A**.

Trap check: B ignores the second transfer, C transfers half of B's initial amount, D compares the final amounts in C and B, and E reports C's final amount rather than the original total.

Question 16

Triangle ABC is right-angled at C , with $AC = 6$, $BC = 8$ and $AB = 10$. The circle with diameter AB is shown. The tangent to the circle at C meets the line AB , extended beyond A , at T .

What is CT ?



A $\frac{60}{7}$

B 12

C 15

D $\frac{120}{7}$

E 20

WORKED SOLUTION

Place $A = (0, 0)$ and $B = (10, 0)$. The 6-8-10 triangle gives $C = (18/5, 24/5)$, and the centre is $O = (5, 0)$. The gradient of OC is

$$\frac{24/5}{18/5 - 5} = -\frac{24}{7}.$$

The tangent is perpendicular to OC , so its gradient is $7/24$. Its equation through C is

$$y - \frac{24}{5} = \frac{7}{24} \left(x - \frac{18}{5} \right).$$

Setting $y = 0$ gives $T = (-90/7, 0)$. Therefore

$$CT = \sqrt{\left(\frac{18}{5} + \frac{90}{7} \right)^2 + \left(\frac{24}{5} \right)^2} = \frac{120}{7}.$$

Therefore the correct option is **D**.

Trap check: The tangent is perpendicular to the radius OC , not to either side of triangle ABC .

Question 17

A right-angled triangle has perpendicular sides **9** and **12**. A square is placed with one side on the hypotenuse and the opposite side parallel to it, with the other two vertices lying on the perpendicular sides.

What is the side length of the square?

A $\frac{180}{37}$

B $\frac{36}{7}$

C $\frac{60}{11}$

D $\frac{72}{13}$

E 6

WORKED SOLUTION

The hypotenuse has length **15**, and the altitude from the right angle to the hypotenuse is

$$h = \frac{2 \times (\frac{1}{2})(9)(12)}{15} = \frac{36}{5}.$$

If the square side is s , the small triangle above it is similar to the original triangle. Its scale factor is $(h - s)/h$, so its base, which is a side of the square, satisfies

$$s = 15 \frac{h - s}{h}.$$

Thus $s(h + 15) = 15h$, and with $h = 36/5$,

$$s = \frac{15(36/5)}{15 + 36/5} = \frac{180}{37}.$$

Therefore the correct option is **A**.

Trap check: This is not the standard square placed at the right-angle vertex; here a side of the square lies on the hypotenuse.

Question 18

A data set contains only the values shown.

Value	1	2	3	4	5
Frequency	2	a	4	b	2

There are **20** observations and their mean is **3.1**. Two observations are selected at random without replacement.

What is the probability that they have equal values?

A $\frac{39}{200}$

B $\frac{39}{190}$

C $\frac{31}{190}$

D $\frac{39}{380}$

E $\frac{3}{10}$

WORKED SOLUTION

The total frequency gives $a + b = 12$. The total of all observations is $20(3.1) = 62$, so

$$2 + 2a + 12 + 4b + 10 = 62 \implies a + 2b = 19.$$

Hence $b = 7$ and $a = 5$. The number of unordered equal-value pairs is

$$\binom{2}{2} + \binom{5}{2} + \binom{4}{2} + \binom{7}{2} + \binom{2}{2} = 1 + 10 + 6 + 21 + 1 = 39.$$

There are $\binom{20}{2} = 190$ unordered pairs in total, so the probability is $39/190$.

Therefore the correct option is B.

Trap check: Using 20^2 as the denominator models selection with replacement and also permits selecting the same observation twice.

Question 19

A price is increased by $p\%$, where $p > 0$. The increased price is then reduced by $(p + 10)\%$. The final price is **12%** below the original price.

What is p ?

A -20

B 12

C 22

D 10

E 20

WORKED SOLUTION

Let $x = p/100$. The two multipliers give

$$(1 + x)(0.9 - x) = 0.88.$$

Expanding and simplifying,

$$0.9 - 0.1x - x^2 = 0.88 \implies x^2 + 0.1x - 0.02 = 0.$$

Hence $(x - 0.1)(x + 0.2) = 0$. Since $p > 0$, $x = 0.1$, so $p = 10$.

Therefore the correct option is D.

Trap check: A is the rejected algebraic root, B copies the final percentage, C adds 12 and 10, and E neglects the product term in the two successive multipliers.

Question 20

What is the smallest positive integer n for which both $72n$ is a perfect square and $50n$ is a perfect cube?

A 20000**B** 5000**C** 10000**D** 16000**E** 40000**WORKED SOLUTION**

Write $n = 2^a 3^b 5^c$ with the least possible non-negative exponents. Since $72 = 2^3 3^2$, the exponents in $72n$ must be even. Since $50 = 2 \cdot 5^2$, the exponents in $50n$ must be multiples of 3. The least simultaneous choices are

$$a = 5, \quad b = 0, \quad c = 4.$$

Thus $n = 2^5 \cdot 5^4 = 32 \cdot 625 = 20000$.

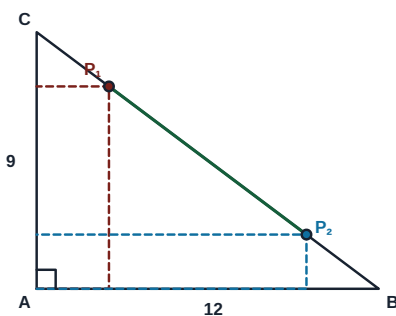
Therefore the correct option is **A**.

Trap check: The same n must satisfy both exponent conditions simultaneously; satisfying only the square condition is insufficient.

Question 21

Triangle ABC is right-angled at A , with $AB = 12$ and $AC = 9$. For a point P on the hypotenuse, perpendiculars from P to AB and AC form a rectangle having A and P as opposite vertices.

There are two positions, P_1 and P_2 , for which the rectangle has area 18. What is P_1P_2 ?

**A** $3\sqrt{3}$ **B** $4\sqrt{3}$ **C** 5**D** $5\sqrt{3}$ **E** $7\sqrt{3}$ **WORKED SOLUTION**

Place $A = (0, 0)$, $B = (12, 0)$ and $C = (0, 9)$. If $P = (x, y)$, the hypotenuse equation is $y = 9 - \frac{3x}{4}$. The rectangle area condition gives

$$x \left(9 - \frac{3}{4}x \right) = 18 \implies x^2 - 12x + 24 = 0.$$

The two roots are $6 \pm 2\sqrt{3}$, so the horizontal separation of P_1, P_2 is $4\sqrt{3}$. Their vertical separation is $(\frac{3}{4})(4\sqrt{3}) = 3\sqrt{3}$. Hence

$$P_1P_2 = \sqrt{(4\sqrt{3})^2 + (3\sqrt{3})^2} = 5\sqrt{3}.$$

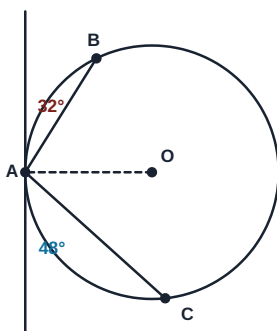
Therefore the correct option is **D**.

Trap check: The horizontal root difference is only $4\sqrt{3}$; P_1P_2 also includes the vertical separation.

Question 22

At a point **A** on a circle, a tangent is drawn. Chords **AB** and **AC** lie on opposite sides of the radius through **A**. The acute angles between the tangent and the chords **AB** and **AC** are 32° and 48° , respectively.

What is the smaller angle **BOC**, where **O** is the centre?



A 100°

B 80°

C 160°

D 200°

E 40°

WORKED SOLUTION

By the tangent–chord theorem,

$$\angle ACB = 32^\circ, \quad \angle ABC = 48^\circ.$$

Therefore $\angle BAC = 180^\circ - 32^\circ - 48^\circ = 100^\circ$. The arc **BC** not containing **A** has central angle 200° . The smaller angle at the centre is therefore

$$360^\circ - 200^\circ = 160^\circ.$$

Therefore the correct option is **C**.

Trap check: A gives the inscribed angle, B adds the two tangent–chord angles without doubling, D selects the reflex central angle, and E doubles their difference.

Question 23

An increasing arithmetic sequence has positive integer terms. Its terms satisfy

$$u_6^2 = u_3 u_{11} \quad \text{and} \quad u_{11} - u_3 = 48.$$

What is the first term of the sequence?

A 6

B 9

C 12

D 18

E 15

WORKED SOLUTION

Let the first term be a and common difference be $d > 0$. Then

$$u_3 = a + 2d, \quad u_6 = a + 5d, \quad u_{11} = a + 10d.$$

The difference condition gives $8d = 48$, so $d = 6$. Expanding the other condition,

$$(a + 5d)^2 = (a + 2d)(a + 10d)$$

$$a^2 + 10ad + 25d^2 = a^2 + 12ad + 20d^2,$$

so $5d^2 = 2ad$. Since $d > 0$, $a = 5d/2 = 15$.

Therefore the correct option is **E**.

Trap check: The identity is multiplicative, not the ordinary arithmetic-mean relation for u_3 , u_6 and u_{11} .

Question 24

Two fair six-sided dice are thrown.

Given that the product of the two scores is divisible by 6, what is the probability that their sum is prime?

A $\frac{7}{15}$

B $\frac{8}{15}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{9}$

WORKED SOLUTION

The 15 ordered outcomes whose product is divisible by 6 are

(1, 6); (2, 3), (2, 6); (3, 2), (3, 4), (3, 6); (4, 3), (4, 6); (5, 6); (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6).

Of these, the eight outcomes

(1, 6), (6, 1), (2, 3), (3, 2), (3, 4), (4, 3), (5, 6), (6, 5)

have prime sum. The required probability is therefore $\frac{8}{15}$.

Therefore the correct option is B.

Trap check: A misses one favourable ordered outcome, C counts unordered pairs, D includes one non-prime sum, and E uses the eight favourable outcomes over the original 36-outcome sample space.

Question 25

The arithmetic sequence

$$5, 12, 19, 26, \dots$$

contains terms that are divisible by **11** but not divisible by **3**. What is the position of the **15th** such term?

A 214

B 225

C 236

D 247

E 258

F 159

WORKED SOLUTION

The n th term is $7n - 2$. Divisibility by **11** requires

$$7n \equiv 2 \pmod{11},$$

so $n \equiv 5 \pmod{11}$. Write $n = 5 + 11k$, where $k \geq 0$. The term is divisible by **3** precisely when $n \equiv 2 \pmod{3}$. Since

$$5 + 11k \equiv 2 + 2k \pmod{3},$$

this happens when $2k \equiv 0 \pmod{3}$, so $k \equiv 0 \pmod{3}$. These values must be excluded. The allowed values of k are therefore

$$1, 2, 4, 5, 7, 8, \dots$$

There are two allowed values in each block of three. The **15th** allowed value is $k = 22$, giving

$$n = 5 + 11(22) = 247.$$

Indeed, $7(247) - 2 = 1727 = 11 \times 157$, which is not divisible by **3**.

Therefore the correct option is D.

Trap check: C excludes the wrong residue class of k ; F counts the 15th multiple-of-11 candidate without any exclusion; A, B and E are one-block or one-candidate counting shifts.

Question 26

A positive real number x satisfies

$$x + \frac{1}{x} = 3.$$

For positive integers n , define

$$A_n = x^n + \frac{1}{x^n}.$$

What is the remainder when A_{2025} is divided by 10?

A 2

B 3

C 5

D 7

E 8

WORKED SOLUTION

Multiplying A_n by $x + x^{-1} = 3$ gives

$$3A_n = A_{n+1} + A_{n-1},$$

so $A_{n+1} = 3A_n - A_{n-1}$. Also $A_0 = 2$ and $A_1 = 3$. Working modulo 10,

$$A_0, A_1, A_2, A_3, A_4, A_5, A_6 \equiv 2, 3, 7, 8, 7, 3, 2.$$

The pair $(A_6, A_7) \equiv (2, 3)$ repeats the initial pair, so the remainders have period 6. Since 2025 leaves remainder 3 on division by 6,

$$A_{2025} \equiv A_3 \equiv 8 \pmod{10}.$$

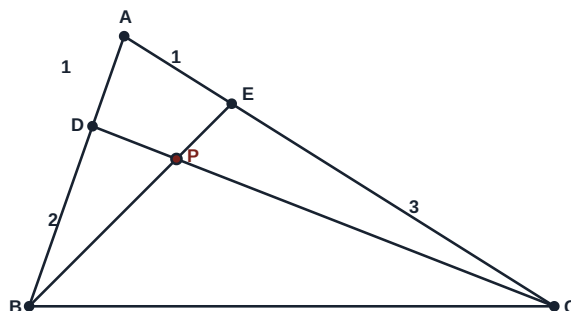
Therefore the correct option is E.

Trap check: Track a pair of consecutive remainders to justify the cycle; matching one term alone is not enough.

Question 27

In triangle ABC , point D lies on AB with $AD : DB = 1 : 2$, and point E lies on AC with $AE : EC = 1 : 3$. Lines DC and EB intersect at P .

What is $EP : PB$?



A 3 : 7

B 8 : 3

C 3 : 8

D 4 : 9

E 2 : 7

F 3 : 11

G 11 : 3

H 8 : 11

WORKED SOLUTION

Choose convenient coordinates $A = (0, 0)$, $B = (3, 0)$, $C = (0, 4)$. Then $D = (1, 0)$ and $E = (0, 1)$. Parameterise P on both cevians:

$$P = D + t(C - D) = (1 - t, 4t),$$

$$P = E + s(B - E) = (3s, 1 - s).$$

Solving $1 - t = 3s$ and $4t = 1 - s$ gives $s = 3/11$. Since s is the fraction of the way from E to B ,

$$EP : PB = s : (1 - s) = \frac{3}{11} : \frac{8}{11} = 3 : 8.$$

Therefore the correct option is **C**.

Trap check: B reverses the required ratio; F reports $EP : EB = 3 : 11$; G reverses that whole-segment ratio; H reports $PB : EB = 8 : 11$. A, D and E arise from using one of the side ratios directly or mismatching coefficients in the two line parameters.

ANSWER KEY

ESAT 2024-25 Recall Mock - Physics

27 hard, recall-informed Physics questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Physics answer key

Q1	E	Q2	B	Q3	A	Q4	C	Q5	E	Q6	B	Q7	C	Q8	A	Q9	B
Q10	D	Q11	C	Q12	E	Q13	C	Q14	C	Q15	A	Q16	C	Q17	E	Q18	B
Q19	B	Q20	C	Q21	D	Q22	E	Q23	C	Q24	B	Q25	D	Q26	A	Q27	E

Physics

27 hard, recall-informed Physics questions with complete worked solutions.

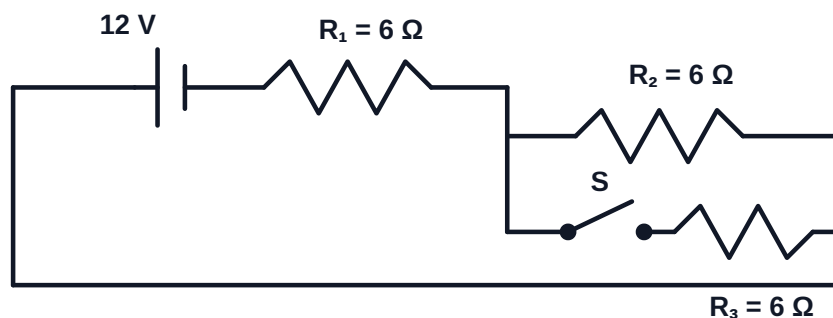
P-01

Qualitative circuit changes

Answer **E**

Question 1

Three identical $6\ \Omega$ resistors are connected to an ideal $12\ \text{V}$ source as shown. Initially switch S is open; it is then closed.



Which statements about closing S are correct?

1. The current supplied by the source increases.
2. The potential difference across R_1 increases.
3. The power dissipated in R_2 decreases.

A 1 only**B** 2 only**C** 1 and 3 only**D** 2 and 3 only**E** 1, 2 and 3

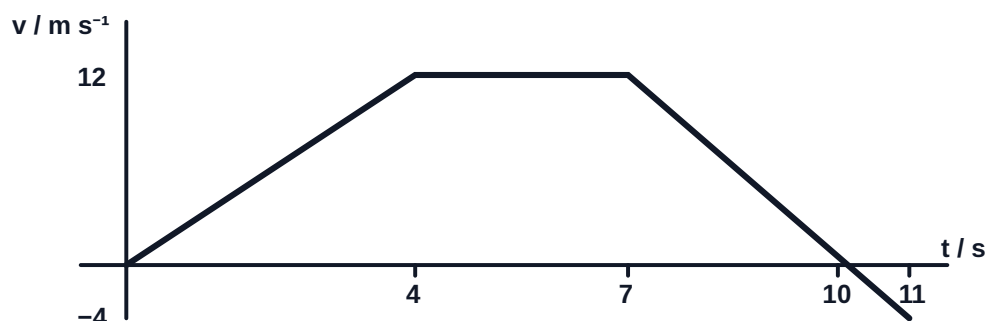
WORKED SOLUTION

With S open, the total resistance is $12\ \Omega$, so the source current is $1\ \text{A}$. The voltage across R_1 is $6\ \text{V}$, and R_2 dissipates $6^2/6 = 6\ \text{W}$.

Closing S puts R_2 and R_3 in parallel: their resistance is $3\ \Omega$, so the total is $9\ \Omega$. The source current rises to $4/3\ \text{A}$, the voltage across R_1 rises to $8\ \text{V}$, and the parallel branch voltage becomes $4\ \text{V}$. Thus R_2 's power falls to $4^2/6 = 8/3\ \text{W}$. All three statements are correct, so the answer is **E**.

Question 2

A particle starts from rest. It accelerates uniformly to 12 m s^{-1} in 4 s , travels at this velocity for 3 s , then changes velocity uniformly to -4 m s^{-1} over the next 4 s . What are its total displacement and total distance travelled?



A 80 m, 80 m

B 76 m, 80 m

C 80 m, 84 m

D 76 m, 76 m

E 72 m, 76 m

WORKED SOLUTION

Signed area gives displacement:

$$\frac{1}{2}(4)(12) + (3)(12) + \frac{1}{2}(12 - 4)(4) = 24 + 36 + 16 = 76 \text{ m.}$$

In the final section the velocity reaches zero after 3 s . Its distance in that section is $\frac{1}{2}(3)(12) + \frac{1}{2}(1)(4) = 20 \text{ m}$. Thus total distance is $24 + 36 + 20 = 80 \text{ m}$. The correct option is **B**.

Question 3

Five complete cycles of a wave pass a fixed point in medium X in 0.020 s . Adjacent crests in X are 0.80 m apart. The wave then enters medium Y obliquely, where its speed is $\frac{3}{4}$ of its speed in X . Which row gives the speed, frequency and wavelength in Y , together with the direction of bending?

A 150 m s^{-1} , 250 Hz , 0.60 m ; towards the normal

B 150 m s^{-1} , 187.5 Hz , 0.80 m ; towards the normal

C 200 m s^{-1} , 250 Hz , 0.60 m ; away from the normal

D 150 m s^{-1} , 250 Hz , 0.60 m ; away from the normal

E 200 m s^{-1} , 250 Hz , 0.80 m ; no change in direction

WORKED SOLUTION

The frequency is $f = 5/0.020 = 250\text{ Hz}$. In X , $v_X = f\lambda_X = 250(0.80) = 200\text{ m s}^{-1}$. Therefore

$$v_Y = \frac{3}{4}v_X = 150\text{ m s}^{-1}, \quad \lambda_Y = \frac{v_Y}{f} = \frac{150}{250} = 0.60\text{ m}.$$

Frequency does not change at the boundary. Since the wave slows down, it bends towards the normal. The correct option is **A**.

Question 4

Two rods P and Q of equal cross-sectional area are joined end to end. Rod P has length L and thermal conductivity k . Rod Q has length $2L$ and thermal conductivity $3k$. The free end of P is kept at 100°C , and the free end of Q at 20°C . Assume one-dimensional conduction: the rods' sides are thermally insulated, contact resistance at the junction is negligible, and no heat is lost to the surroundings. In steady state, the transfer rate through a rod is $kA\Delta T/L$. What is the junction temperature?

A 36°C

B 44°C

C 52°C

D 60°C

E 68°C

F 48°C

WORKED SOLUTION

The thermal resistances are proportional to $L/(kA)$. Thus

$$R_P : R_Q = 1 : \frac{2}{3} = 3 : 2.$$

The total temperature drop is 80°C , so the drop across P is $\frac{3}{5}(80) = 48^\circ\text{C}$. The junction is therefore at $100 - 48 = 52^\circ\text{C}$. The correct option is **C**.

Question 5

A simple AC generator contains a coil of N turns rotating at angular speed ω in a uniform magnetic field and is connected to a fixed resistor. For this generator, $V_{\text{peak}} \propto N\omega$, $f \propto \omega$, and the average power in the resistor is proportional to V_{peak}^2 .

The coil is replaced by one of $2N$ turns rotating at angular speed $\omega/2$, with all other factors unchanged. How do the peak voltage and the energy delivered to the resistor in one complete cycle change?

A Peak voltage doubles; energy per cycle doubles

B Peak voltage is unchanged; energy per cycle is unchanged

C Peak voltage halves; energy per cycle halves

D Peak voltage doubles; energy per cycle is four times as large

E Peak voltage is unchanged; energy per cycle doubles

WORKED SOLUTION

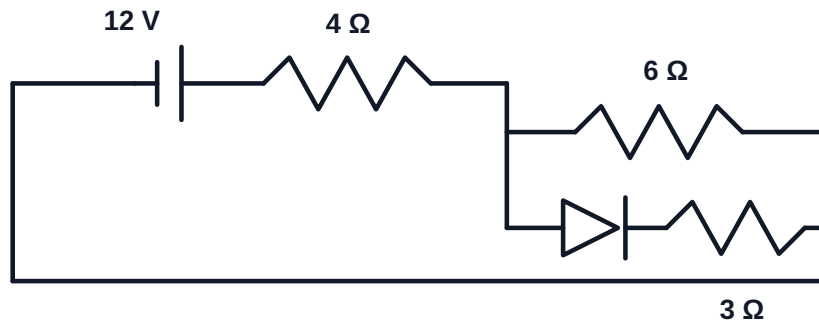
The peak induced voltage scales as $N\omega$, so the two changes multiply it by $2 \times \frac{1}{2} = 1$: the peak voltage is unchanged. The average power in the fixed resistor is therefore unchanged.

The frequency halves, so the period of one cycle doubles. Since energy per cycle equals average power multiplied by the period, the energy delivered in each complete cycle doubles. The correct option is **E**.

Question 6

An ideal **12 V** source supplies a **4 Ω** resistor in series with two parallel branches. One branch is **6 Ω**; the other contains an ideal diode in series with **3 Ω**. With the first polarity the diode conducts. When the source polarity is reversed, the diode blocks. What is the ratio

$$\frac{\text{power drawn with the diode conducting}}{\text{power drawn with the diode blocking}}?$$



A $\frac{2}{3}$

B $\frac{5}{3}$

C $\frac{3}{2}$

D $\frac{5}{2}$

E $\frac{10}{3}$

WORKED SOLUTION

When the diode conducts, **6 Ω** and **3 Ω** are in parallel, giving **2 Ω**. Total resistance is **6 Ω**, so $P_{\text{on}} = V^2/R = 144/6 = 24 \text{ W}$.

When it blocks, total resistance is **4 + 6 = 10 Ω**, so $P_{\text{off}} = 144/10 = 14.4 \text{ W}$. Therefore $P_{\text{on}}/P_{\text{off}} = 24/14.4 = 5/3$. The correct option is **B**.

Question 7

A Geiger counter records the following counts in repeated **30 s** intervals. The source and detector remain fixed.

Measurement	1	2	3	4
Source + background	148	161	153	139
Background only	18	22	16	24

Which statements are correct?

1. Repeated counts can differ even if the source activity and apparatus are unchanged.
2. The best estimate from these data of the source-only count in **30 s** is **130.25**.
3. Subtracting the mean background removes all random uncertainty from the source estimate.

A 1 only

B 2 only

C 1 and 2 only

D 2 and 3 only

E 1, 2 and 3

WORKED SOLUTION

Radioactive decay is random, so statement 1 is correct. The mean source-plus-background count is $(148 + 161 + 153 + 139)/4 = 150.25$, while the mean background is $(18 + 22 + 16 + 24)/4 = 20$. Their difference is **130.25**, so statement 2 is correct.

Background subtraction corrects the central estimate but does not eliminate fluctuations in either set of counts, so statement 3 is false. The correct option is **C**.

Question 8

A **1200 kg** car pulls an **800 kg** trailer along a horizontal road. The resistive forces on the car and trailer are **400 N** and **200 N**, respectively. Both accelerate forwards at **1.5 m s^{-2}** . What are the driving force on the car and the force exerted by the trailer on the car?

A 3600 N forwards; 1400 N backwards

B 3000 N forwards; 1200 N backwards

C 3600 N forwards; 1200 N backwards

D 3400 N forwards; 1400 N backwards

E 3600 N forwards; 1400 N forwards

WORKED SOLUTION

For the complete car-trailer system, if the driving force is F , then

$$F - (400 + 200) = (1200 + 800)(1.5),$$

so $F = 3600 \text{ N}$. For the trailer alone, if the coupling force is T ,

$$T - 200 = 800(1.5),$$

giving $T = 1400 \text{ N}$. By Newton's third law, the trailer exerts an equal **1400 N** force backwards on the car. The correct option is **A**.

Question 9

A sound source emits **240 Hz** while moving directly towards a stationary observer at **60 m s^{-1}** . The speed of sound in the still air is **360 m s^{-1}** . During one source period, a crest travels at the speed of sound while the source moves closer to the preceding crest. What frequency does the observer detect?

A 270 Hz

B 288 Hz

C 240 Hz

D 300 Hz

E 320 Hz

WORKED SOLUTION

One source period is $T = 1/240 \text{ s}$. In this time the earlier crest travels $360T$, while the source advances $60T$. The wavelength in front of the source is therefore

$$\lambda = (360 - 60)T = \frac{300}{240} = 1.25 \text{ m}.$$

The stationary observer receives waves travelling at **360 m s^{-1}** , so

$$f_{\text{obs}} = \frac{360}{1.25} = 288 \text{ Hz}.$$

The correct option is **B**.

Question 10

A transformer has **240 V** across its primary and **12 V** across its secondary. A **6 Ω** resistor is connected across the secondary. The transformer is 80% efficient, where efficiency is secondary output power divided by primary input power. Assume that its terminal voltage ratio still obeys $V_s/V_p = N_s/N_p$. What are the primary current I_p and the turns ratio N_s/N_p ?

A $I_p = \frac{1}{10} \text{ A}, N_s/N_p = \frac{1}{20}$

B $I_p = \frac{1}{8} \text{ A}, N_s/N_p = \frac{1}{10}$

C $I_p = \frac{1}{10} \text{ A}, N_s/N_p = \frac{1}{10}$

D $I_p = \frac{1}{8} \text{ A}, N_s/N_p = \frac{1}{20}$

E $I_p = \frac{1}{5} \text{ A}, N_s/N_p = \frac{1}{20}$

WORKED SOLUTION

The secondary current is $12/6 = 2 \text{ A}$, so output power is $12 \times 2 = 24 \text{ W}$. Input power is $24/0.80 = 30 \text{ W}$, giving $I_p = 30/240 = 1/8 \text{ A}$. Also

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{12}{240} = \frac{1}{20}.$$

The correct option is **D**.

Question 11

A fixed mass of ideal gas initially has pressure p , volume V , absolute temperature T and density ρ . It is first compressed isothermally to volume $V/2$. Keeping this new volume fixed, it is then heated until its absolute temperature is $3T/2$. What are its final pressure and density? For this question, use $pV/T = \text{constant}$ for a fixed mass of ideal gas.

A $\frac{3}{2}p, 2\rho$

B $2p, \frac{3}{2}\rho$

C $3p, 2\rho$

D $3p, 3\rho$

E $4p, 2\rho$

WORKED SOLUTION

During the isothermal compression, pV is constant. Halving the volume therefore doubles the pressure to $2p$. At fixed volume, pressure is proportional to absolute temperature, so the second step multiplies the pressure by $3/2$:

$$p_{\text{final}} = 2p \left(\frac{3}{2} \right) = 3p.$$

The mass is unchanged and the final volume is $V/2$, so $\rho_{\text{final}} = m/(V/2) = 2\rho$. The correct option is **C**.

Question 12

An object falls vertically through a fluid. Its weight is W , its upthrust is a constant U , and its drag force is kv^2 , where v is its speed. Its terminal speed is v_t . What is the resultant force when its speed is $v_t/2$?

A $\frac{1}{4}(W - U)$ downward

B $\frac{1}{2}(W - U)$ downward

C $\frac{3}{4}(W - U)$ upward

D $\frac{1}{4}(W - U)$ upward

E $\frac{3}{4}(W - U)$ downward

WORKED SOLUTION

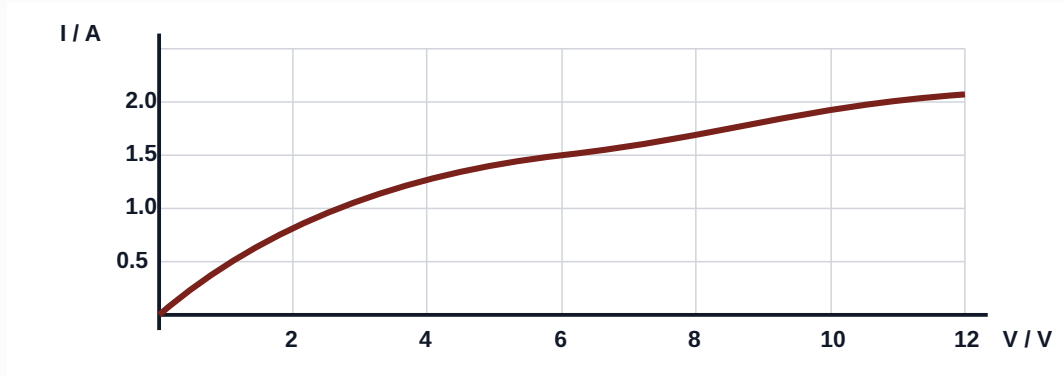
At terminal speed, $kv_t^2 = W - U$. At $v_t/2$, the drag is $\frac{1}{4}kv_t^2 = \frac{1}{4}(W - U)$. The downward resultant is therefore

$$(W - U) - \frac{1}{4}(W - U) = \frac{3}{4}(W - U).$$

The correct option is **E**.

Question 13

The graph shows the current-voltage characteristic of a filament lamp. The lamp is connected in series with a 4Ω resistor across an ideal 12 V source. What power is dissipated in the lamp?

**A** 4.5 W**B** 6 W**C** 9 W**D** 12 W**E** 18 W**WORKED SOLUTION**

If the lamp voltage is V_L and the current is I , the series resistor has voltage $4I$, so

$$V_L + 4I = 12.$$

From the graph, the lamp has $I = 1.5\text{ A}$ at $V_L = 6\text{ V}$; this satisfies $6 + 4(1.5) = 12$. The increasing lamp curve and decreasing load line give a unique operating point. Thus $P_L = V_L I = 6(1.5) = 9\text{ W}$. The correct option is **C**.

Question 14

Equal masses of solids **A** and **B**, both initially at **20°C**, are heated by identical constant-power heaters. Energy loss is negligible. The results are:

Solid	Melting temperature	Time to reach melting temperature	Time at constant temperature while melting
A	60°C	40 s	50 s
B	80°C	60 s	30 s

Which statements are correct?

1. Solid B has the higher melting point.
2. In the solid state, A has the greater specific heat capacity.
3. A has the greater specific latent heat of fusion.

A 1 only

B 2 only

C 1 and 3 only

D 2 and 3 only

E 1, 2 and 3

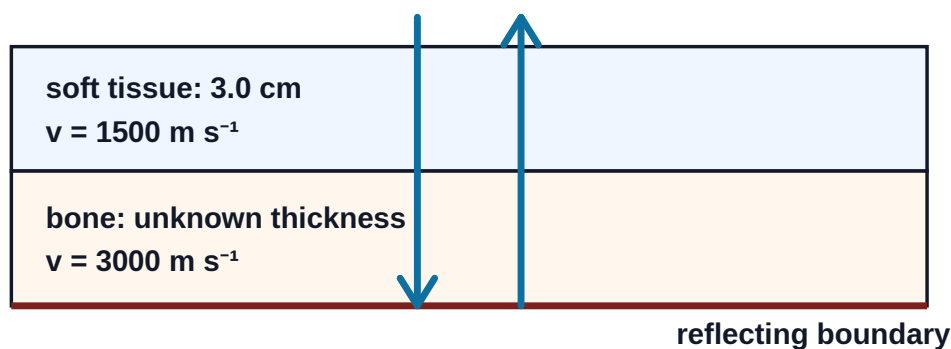
WORKED SOLUTION

The melting temperatures make statement 1 true. Before melting, $Pt = mc\Delta T$. For A, $t/\Delta T = 40/40 = 1 \text{ s K}^{-1}$; for B, $60/60 = 1 \text{ s K}^{-1}$. The masses and powers are equal, so the specific heat capacities are equal and statement 2 is false.

During melting, $Pt = mL$. A remains on the plateau for longer, so A has the greater latent heat and statement 3 is true. The correct option is **C**.

Question 15

An ultrasound pulse travels normally through **3.0 cm** of soft tissue at **1500 m s^{-1}** , then through bone at **3000 m s^{-1}** . It reflects from a boundary within the bone and returns along the same path. The echo is received **$60 \mu\text{s}$** after emission. How far below the surface is the reflecting boundary?

**A** 6.0 cm**B** 3.0 cm**C** 4.5 cm**D** 7.5 cm**E** 9.0 cm**WORKED SOLUTION**

The round-trip time in the soft tissue is

$$t_s = \frac{2(0.030)}{1500} = 40 \mu\text{s}.$$

The remaining **$20 \mu\text{s}$** is the round-trip time in bone. If the one-way bone thickness is **x** ,

$$\frac{2x}{3000} = 20 \times 10^{-6} \Rightarrow x = 0.030 \text{ m} = 3.0 \text{ cm}.$$

The reflecting boundary is therefore **$3.0 + 3.0 = 6.0 \text{ cm}$** below the surface. The correct option is **A**.

Question 16

An electric pump takes in power at **500 W** and is **80%** efficient. It pumps water at **2.0 kg s⁻¹** from a large reservoir to an outlet **15 m** higher. The reservoir surface and outlet are both at atmospheric pressure, and the speed of the reservoir surface is negligible. Neglect all losses other than those represented by the efficiency, and take **$g = 10 \text{ N kg}^{-1}$** .

What is the speed of the water as it leaves the outlet?

A 5 m s^{-1}

B $5\sqrt{2} \text{ m s}^{-1}$

C 10 m s^{-1}

D $10\sqrt{2} \text{ m s}^{-1}$

E 20 m s^{-1}

WORKED SOLUTION

The useful power is $0.80(500) = 400 \text{ W}$. Raising **2.0 kg** each second through **15 m** requires gravitational power

$$\dot{m}gh = 2(10)(15) = 300 \text{ W}.$$

The remaining **100 W** becomes kinetic energy of the water each second:

$$\frac{1}{2}\dot{m}v^2 = 100 \Rightarrow \frac{1}{2}(2)v^2 = 100.$$

Thus **$v = 10 \text{ m s}^{-1}$** . The correct option is **C**.

Question 17

The north pole of a bar magnet moves directly towards one end of a coil at speed **u** , is stopped, and is then pulled away. As it passes the same positions on the way out, its speed is **$u/2$** . Which statements are correct?

1. While the north pole approaches, the near face of the coil behaves as a north pole.
2. While the magnet is stationary, the induced potential difference is zero.
3. While the magnet recedes through the same positions, the induced potential difference has the opposite direction and a smaller magnitude than during approach.

A 1 only

B 2 only

C 1 and 2 only

D 2 and 3 only

E 1, 2 and 3

WORKED SOLUTION

Lenz's law makes the near face repel the approaching north pole, so it acts north. No flux change occurs while the magnet is stationary. On withdrawal, the flux change reverses, reversing the induced voltage; at the same position its rate of change is smaller because the speed is halved. All statements are correct, so the answer is **E**.

Question 18

A source emits alpha, beta-minus and gamma radiation. A detector has a background rate of **20 counts min⁻¹**. It records:

- **620 counts min⁻¹** with no absorber;
- **420 counts min⁻¹** after paper is inserted;
- **170 counts min⁻¹** after aluminium is added behind the paper;
- **20 counts min⁻¹** after thick lead is added.

Assume the paper removes only alpha, the aluminium then removes only beta-minus, and the lead removes the remaining gamma. Which option gives, respectively, the ratio of the source's beta-minus count rate to its gamma count rate and the fraction of the source-only count rate due to alpha radiation?

A 2 : 1; $\frac{1}{3}$

B 5 : 3; $\frac{1}{3}$

C 3 : 2; $\frac{10}{31}$

D 5 : 4; $\frac{1}{2}$

E 3 : 5; $\frac{1}{3}$

WORKED SOLUTION

The beta-minus component is the decrease when aluminium is added:

$$420 - 170 = 250 \text{ counts min}^{-1}.$$

After paper and aluminium, the source contribution is the gamma rate. Subtracting background gives

$$170 - 20 = 150 \text{ counts min}^{-1}.$$

The alpha component is the decrease when paper is inserted, $620 - 420 = 200 \text{ counts min}^{-1}$. The total source-only rate is $620 - 20 = 600 \text{ counts min}^{-1}$, so the alpha fraction is $200/600 = 1/3$.

Thus the required pair is **5 : 3; 1/3**. The correct option is **B**.

Question 19

A vertical-sided container of base area A holds an oil layer of density ρ and depth $2h$ above a liquid of density 2ρ and depth h . The same volumes of both liquids are transferred to a second vertical-sided container of base area $2A$. Compared with the first container, how do the gauge pressure and the total force due to liquid pressure on the horizontal base change?

- A** Pressure halves; force halves
- B** Pressure halves; force is unchanged
- C** Pressure is unchanged; force doubles
- D** Pressure becomes one quarter; force halves
- E** Pressure doubles; force is unchanged

WORKED SOLUTION

Initially, $p_1 = \rho g(2h) + 2\rho g(h) = 4\rho gh$. Doubling base area halves each depth, so

$$p_2 = \rho g(h) + 2\rho g\left(\frac{h}{2}\right) = 2\rho gh = \frac{p_1}{2}.$$

But $F_1 = p_1 A = 4\rho ghA$, while $F_2 = p_2(2A) = 4\rho ghA$. Pressure halves and force is unchanged. The correct option is **B**.

Question 20

A positively charged rod is brought close to an initially neutral isolated metal sphere without touching it. While the rod remains nearby, the sphere is briefly connected to Earth. The Earth connection is removed before the rod is taken away. What is the final charge on the sphere, and what happened while it was earthed?

- A** Neutral; equal numbers of electrons flowed in both directions
- B** Positive; electrons flowed from Earth to the sphere
- C** Negative; electrons flowed from Earth to the sphere
- D** Positive; electrons flowed from the sphere to Earth
- E** Negative; positive charge flowed from Earth to the sphere

WORKED SOLUTION

The positive rod attracts electrons towards the sphere. While earthed, additional electrons flow from Earth onto it. Removing the Earth connection traps this excess; after the rod is removed, the electrons redistribute and the sphere remains negative. The correct option is **C**.

Question 21

Two otherwise identical hot metal containers are at the same temperature in identical cooler surroundings. The initial net radiative loss from the dull black container is four times that from the polished silver container. Each also loses energy by non-radiative processes at the same rate H . The measured initial total loss rates are **70 W** for the black container and **40 W** for the silver container. What fraction of the black container's total loss is by radiation?

A $\frac{3}{7}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{4}{7}$

E $\frac{5}{7}$

WORKED SOLUTION

Let the black container's radiative loss be R . The silver container's radiative loss is then $R/4$. Hence

$$R + H = 70, \quad \frac{R}{4} + H = 40.$$

Subtracting gives $3R/4 = 30$, so $R = 40 \text{ W}$. The required fraction is therefore

$$\frac{R}{70} = \frac{40}{70} = \frac{4}{7}.$$

The correct option is **D**.

Question 22

A particle of mass m travels at speed $3u$. It collides with a particle of mass $2m$ travelling in the same direction at speed u . They stick together. Which option gives their common speed and the fraction of the initial kinetic energy that is lost?

A $\frac{5u}{3}; \frac{8}{25}$

B $\frac{4u}{3}; \frac{1}{4}$

C $\frac{5u}{3}; \frac{4}{33}$

D $\frac{4u}{3}; \frac{1}{3}$

E $\frac{5u}{3}; \frac{8}{33}$

WORKED SOLUTION

Momentum conservation gives $3mu + 2mu = 3mv$, so $v = 5u/3$. Initial kinetic energy is $\frac{1}{2}m(3u)^2 + \frac{1}{2}(2m)u^2 = \frac{11}{2}mu^2$. Final kinetic energy is

$$\frac{1}{2}(3m)\left(\frac{5u}{3}\right)^2 = \frac{25}{6}mu^2.$$

The loss is $\frac{4}{3}mu^2$, so the fraction lost is $(4/3)/(11/2) = 8/33$. The required pair is therefore **5u/3; 8/33**, and the correct option is **E**.

Question 23

In medium X , a wave has speed v , wavelength λ and period T . It enters normally into a slab of medium Y , where its speed is $3v/4$. The slab thickness is 6λ . How many wavelengths fit across the slab, and how long does the wave take to cross it?

A 6 wavelengths; $6T$

B 8 wavelengths; $6T$

C 8 wavelengths; $8T$

D 6 wavelengths; $8T$

E 9 wavelengths; $8T$

WORKED SOLUTION

Frequency is unchanged, so the wavelength in Y is $\lambda_Y = (3/4)\lambda$. Therefore

$$\frac{6\lambda}{(3/4)\lambda} = 8$$

wavelengths fit in the slab. Crossing time is $6\lambda/(3v/4) = 8\lambda/v = 8T$, because $T = \lambda/v$. The correct option is **C**.

Question 24

In an original transmission system, power P enters cables of total resistance R at voltage V . The power lost in the cables is exactly $P/5$.

In a redesigned system, the same generator power P enters a transformer of efficiency **80%**, whose output sends power through the same cables at voltage $4V$. At the far end, a second transformer of efficiency **75%** supplies a load.

What fraction of P reaches the load?

A $\frac{3}{5}$

B $\frac{297}{500}$

C $\frac{99}{125}$

D $\frac{12}{25}$

E $\frac{99}{100}$

WORKED SOLUTION

Originally $I = P/V$ and $I^2 R = P/5$. The first transformer sends $4P/5$ into the cables at $4V$, so the new cable current is

$$I' = \frac{4P/5}{4V} = \frac{P}{5V} = \frac{I}{5}.$$

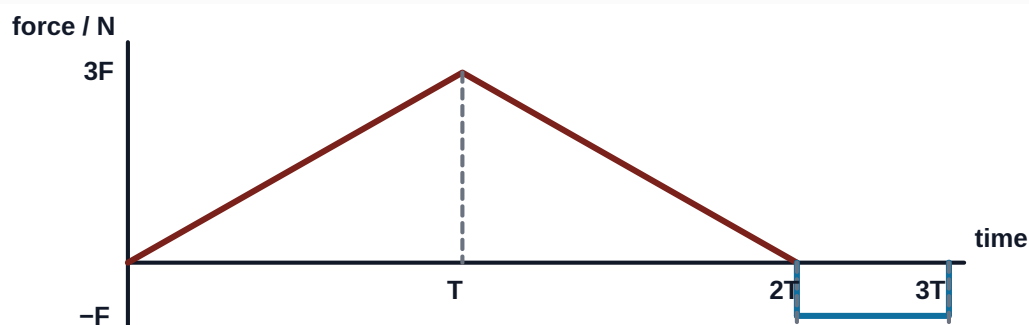
The new cable loss is therefore $(I/5)^2 R = (P/5)/25 = P/125$. The power entering the receiving transformer is

$$\frac{4P}{5} - \frac{P}{125} = \frac{99P}{125}.$$

Its **75%** efficiency gives load power $(3/4)(99P/125) = 297P/500$. The correct option is **B**.

Question 25

A trolley of mass m is initially at rest. The horizontal force on it varies with time as shown. Immediately after $t = 3T$, it collides with a stationary trolley of mass $3m$; the trolleys stick together. They then move up a smooth ramp and come momentarily to rest at vertical height h .



Which expression gives h ?

A $\frac{F^2 T^2}{2m^2 g}$

B $\frac{F^2 T^2}{4m^2 g}$

C $\frac{F^2 T^2}{32m^2 g}$

D $\frac{F^2 T^2}{8m^2 g}$

E $\frac{9F^2 T^2}{32m^2 g}$

WORKED SOLUTION

The positive triangular area is $\frac{1}{2}(2T)(3F) = 3FT$; the negative rectangular area is $-FT$. The net impulse is therefore $2FT$, so the first trolley's momentum just before collision is $2FT$.

Momentum conservation in the sticking collision gives the combined mass $4m$ speed

$$v = \frac{2FT}{4m} = \frac{FT}{2m}.$$

On the smooth ramp, $\frac{1}{2}(4m)v^2 = (4m)gh$, so $h = v^2/(2g) = F^2 T^2/(8m^2 g)$. The correct option is **D**.

Question 26

A lift of mass m is moving vertically upward. At three instants X , Y and Z , its upward speed has the same value v . Its acceleration is upward with magnitude a at X , zero at Y , and downward with magnitude a at Z , where $0 < a < g$. The cable remains taut and resistive forces are negligible.

Which ordering is correct for the instantaneous powers P_X, P_Y, P_Z delivered to the lift by the cable?

A $P_X > P_Y > P_Z$

B $P_Z > P_Y > P_X$

C $P_X = P_Y = P_Z$

D $P_X > P_Z > P_Y$

E $P_Y > P_X > P_Z$

WORKED SOLUTION

Newton's second law gives cable tensions $m(g + a)$, mg and $m(g - a)$ at X , Y and Z , respectively. At each instant the cable force and velocity are parallel and the speed is the same, so $P = Tv$.

Therefore $P_X > P_Y > P_Z$, and the correct option is **A**.

Question 27

A nucleus initially has mass number A , proton number Z and neutron number $N = A - Z$. It emits some alpha particles and some beta-minus particles. Its final mass number is $A - 12$, and its final neutron number is $N - 8$. How many emissions of each type occurred, and what is the final proton number?

A 3 alpha, 2 beta-minus; $Z - 8$

B 3 alpha, 4 beta-minus; $Z - 2$

C 2 alpha, 2 beta-minus; $Z - 2$

D 3 alpha, 2 beta-minus; $Z - 6$

E 3 alpha, 2 beta-minus; $Z - 4$

WORKED SOLUTION

Only alpha decay changes mass number, by -4 each time. A change from A to $A - 12$ therefore requires three alpha decays. These reduce neutron number by 2 each, taking N to $N - 6$.

Each beta-minus decay converts a neutron into a proton, so it reduces neutron number by 1 . Reaching $N - 8$ therefore requires two beta-minus decays. The proton number changes by

$$-2(3) + 1(2) = -4,$$

so the final proton number is $Z - 4$. The correct option is **E**.

ANSWER KEY

ESAT 2024-25 Recall Mock - Biology

27 hard, recall-informed Biology questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Biology answer key

Q1	A	Q2	F	Q3	B	Q4	C	Q5	A	Q6	E	Q7	B	Q8	C	Q9	C
Q10	F	Q11	C	Q12	D	Q13	D	Q14	C	Q15	D	Q16	B	Q17	E	Q18	D
Q19	A	Q20	B	Q21	H	Q22	A	Q23	G	Q24	E	Q25	B	Q26	E	Q27	E

Biology

27 hard, recall-informed Biology questions with complete worked solutions.

B-01

Cells and membrane transport

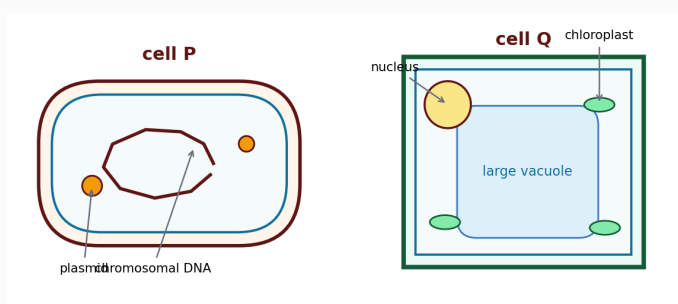
Approachable

Answer **A**

Question 1

The diagrams show two cells, P and Q.

Which conclusion is supported by the diagrams and established cell biology?



Programmatically generated diagram; not necessarily to scale.

- A** P can respire without mitochondria and may contain plasmid DNA.
- B** P cannot release energy because it has no mitochondria.
- C** The cell wall of Q proves that Q cannot be a bacterial cell.
- D** P has its chromosomal DNA enclosed by a nuclear membrane.
- E** The presence of cytoplasm distinguishes Q from P.

WORKED SOLUTION

P is a bacterium. Bacteria carry out respiration using enzymes associated with their cell membrane and cytoplasm, despite lacking mitochondria, and they may contain both chromosomal and plasmid DNA. A cell wall does not uniquely identify a plant cell because bacterial cells also possess walls; both cells also contain cytoplasm.

The correct option is **A**.

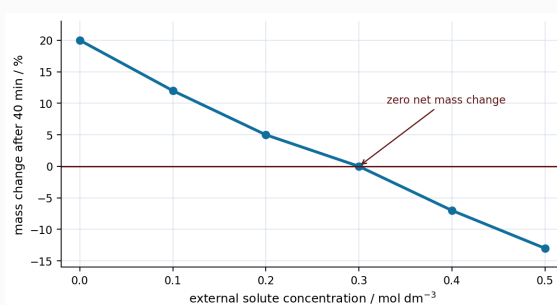
Distractor check: B treats mitochondria as essential for all respiration; C treats a wall as uniquely plant-like; D invents a bacterial nucleus; E overlooks that both cells contain cytoplasm.

Question 2

Identical plant-tissue cylinders are left for 40 minutes in solutions of different solute concentration. The graph shows their mean percentage change in mass. Assume that the cylinders have reached osmotic equilibrium by 40 minutes and that the solute does not cross cell membranes.

1. At 0.30 mol dm^{-3} , zero net mass change indicates that the tissue and the solution have the same water potential.
2. The graph shows that the instantaneous rate of osmosis at 40 minutes is greatest in 0.50 mol dm^{-3} solution.
3. A cylinder with initial mass 4.00 g would have final mass 3.48 g after a 13% decrease.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Under the stated equilibrium and impermeant-solute assumptions, zero net mass change identifies the isotonic concentration, so statement 1 is correct. The graph gives total mass change after 40 minutes, not the instantaneous rate at that time, so statement 2 does not follow. For statement 3, $4.00 \times 0.87 = 3.48 \text{ g}$.

Statements 1 and 3 only are correct, so the answer is **F**.

Distractor check: A large final mass change is not a measurement of the instantaneous rate at the finishing time.

Question 3

A diploid cell has $2n = 8$. In normal meiosis I, homologous chromosomes separate; sister chromatids separate during meiosis II. Before meiosis, the cell's DNA replicates normally. During meiosis I, one homologous pair fails to separate: both homologues move to the same daughter cell. Meiosis II then proceeds normally.

Which outcome is expected?

- A** four gametes with 4 chromosomes each; 16 chromosomes in total
- B** two gametes with 5 chromosomes and two with 3; 16 chromosomes in total
- C** two gametes with 5 chromosomes and two with 3; 8 chromosomes in total
- D** one gamete with 5 chromosomes, one with 3 and two with 4; 16 chromosomes in total
- E** four gametes with 8 chromosomes each; 32 chromosomes in total

WORKED SOLUTION

The normal haploid number is 4. Failure of one homologous pair to separate at meiosis I gives one cell an extra chromosome and the other one fewer. Normal meiosis II therefore produces two gametes with 5 chromosomes and two with 3. Their total is $2(5)+2(3)=16$.

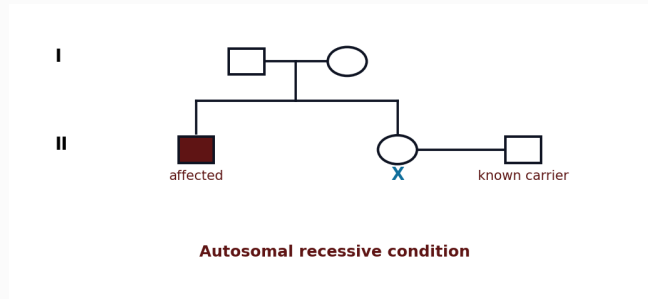
The correct option is **B**.

Distractor check: A describes normal meiosis; D is the pattern expected from a different separation error affecting only one meiosis-II division.

Question 4

The pedigree concerns an autosomal recessive condition. The two unaffected people in generation I have one affected child and one unaffected child, X. X has a partner who is known to be heterozygous.

What is the probability that *exactly one* of the first two children of X and the partner is affected?



Programmatically generated diagram; not necessarily to scale.

A 1/8

B 3/16

C 1/4

D 3/8

E 1/2

WORKED SOLUTION

The generation-I parents must both be carriers. Given that X is unaffected, X has conditional probability $2/3$ of being a carrier. If X is a carrier, the cross is $Aa \times Aa$, so the probability of exactly one affected child in two is

$$2 \times (1/4) \times (3/4) = 3/8.$$

Multiplying by the probability that X is a carrier gives $(2/3)(3/8) = 1/4$. The correct option is **C**.

Distractor check: D assumes X is certainly a carrier; B uses an unjustified one-half carrier probability.

Question 5

One strand of a double-stranded DNA molecule contains 60 nucleotides:

base	A	T	G	C
number on this strand	12	9	15	24

Treat the strand shown as the coding strand, read from its first base in one continuous reading frame. Assume that the 60 bases form complete triplets and that none is a stop triplet. Which option gives:

- the percentage of all nucleotides in the double-stranded molecule that are adenine; and
- the maximum number of amino acids for which this section can code?

A 17.5%; 20**B** 35%; 20**C** 17.5%; 60**D** 20%; 20**E** 35%; 40**WORKED SOLUTION**

The complementary strand contains 9 adenines, paired with the 9 thymines shown. Total adenine is therefore $12+9=21$ out of 120 nucleotides, or 17.5%. Under the stated coding-strand, reading-frame and no-stop assumptions, 60 bases form $60/3=20$ codons, so the section codes for 20 amino acids.

The correct option is **A**.

Distractor check: 20% uses adenine on the shown strand only; 35% incorrectly combines A and T and calls both adenine.

Question 6

A small part of a genetic code is supplied.

triplet	amino acid	triplet	amino acid
AAA	lysine	AAG	lysine
AAC	asparagine	GAA	glutamate
GAG	glutamate	GAC	aspartate

The original sequence is **AAA - GAA - AAC**. Three mutants have sequences:

P: AAG - GAA - AAC

Q: AAA - GAG - AAC

R: AAA - GAA - GAC

1. P specifies the same amino-acid sequence as the original.
2. Q specifies the same amino-acid sequence as the original.
3. R produces the same phenotype as the original.

Which statements are supported by the information supplied?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

AAG and AAA both specify lysine, while GAG and GAA both specify glutamate. P and Q therefore specify the same three amino acids as the original. R changes asparagine to aspartate, and the code table gives no phenotype evidence, so statement 3 is not supported.

Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: A base substitution can be silent, while an amino-acid change cannot by itself establish the resulting phenotype.

Question 7

Which statement correctly compares stem-cell potential with a possible consequence of repeated selective breeding?

- A** Pluripotent cells can form a complete organism, and selective breeding necessarily increases genetic variation.
- B** Totipotent cells can potentially form a complete organism, and repeated selection can reduce genetic variation.
- C** Totipotent and pluripotent cells have identical potential, and selection leaves allele frequencies unchanged.
- D** Only pluripotent cells can form specialised cells, and selective breeding always creates new alleles.
- E** Totipotent cells form fewer cell types than pluripotent cells, and selection prevents inheritance.

WORKED SOLUTION

Totipotent cells can generate all cell types required for a complete organism. Pluripotent cells can form many body cell types but do not have the same complete developmental potential. Repeated selective breeding changes allele frequencies and may remove alleles not associated with the selected phenotype, reducing variation.

The correct option is **B**.

Distractor check: The alternatives reverse totipotent and pluripotent potential or claim that selection necessarily creates variation rather than sometimes reducing it.

Question 8

Before antibiotic treatment, a bacterial population contains 20 000 cells. Of these, 5% carry an inherited resistance variant. The treatment kills 99% of susceptible cells and 10% of resistant cells.

What fraction of the surviving population carries the resistance variant?

- A** 1/20
- B** 9/200
- C** 90/109
- D** 99/100
- E** 18/19

WORKED SOLUTION

There are initially 1000 resistant and 19 000 susceptible cells. Survivors are $0.90(1000)=900$ resistant and $0.01(19000)=190$ susceptible. The required fraction is

$$900/(900+190) = 90/109.$$

The correct option is **C**.

Distractor check: 1/20 is the initial proportion; 18/19 ignores the different survival rate of susceptible cells.

Question 9

Two enzymes are tested at the same temperature with substrate in excess.

pH	4	5	6	7	8
initial rate of A	4	12	16	16	6
initial rate of B	2	6	12	20	10

1. The equal measured rates at pH 6 and pH 7 identify pH 6.5 as the optimum pH of A.
2. At pH 7, the rate for B is 25% greater than the rate for A.
3. The higher rate of B at pH 8 than at pH 4 shows that B retains its normal three-dimensional shape at pH 8.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Equal rates at the two tested pH values do not locate the optimum between them, so statement 1 is not supported. At pH 7, the increase from 16 to 20 is $\frac{4}{16}=25\%$, so statement 2 is correct. A non-zero rate does not show that every enzyme molecule retains its normal shape, so statement 3 does not follow.

Statement 2 only is correct, so the answer is **C**.

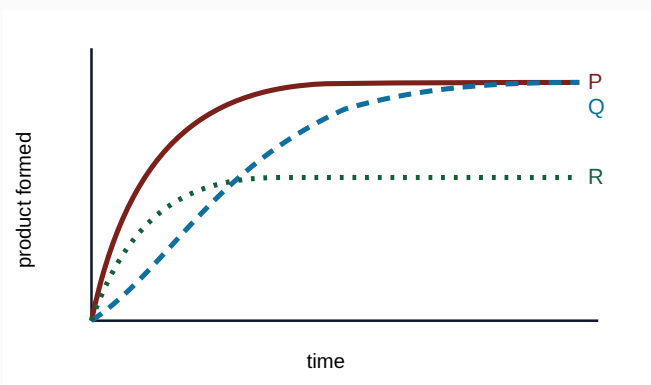
Distractor check: Discrete measurements cannot locate an untested optimum, and residual activity is not evidence about every enzyme molecule.

Question 10

Three experiments use identical initial amounts of enzyme and substrate. Only temperature differs. The graph shows product formed.

1. P has a greater initial rate than Q.
2. The common final product amount for P and Q shows that enzyme molecules remained structurally unchanged in both experiments.
3. R is consistent with enzyme becoming inactive before all the substrate is converted.

Which statements are correct?



A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Initial rate is the initial gradient, which is steeper for P than for Q. Equal final amounts do not show that every enzyme molecule remained unchanged; both experiments can still eventually convert the same finite amount of substrate. R rises rapidly but stops at a lower product amount despite the identical start, consistent with loss of functioning active sites before completion.

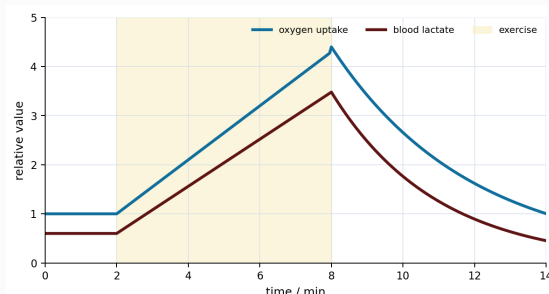
Statements 1 and 3 only are correct, so the answer is **F**.

Distractor check: A plateau records product amount, not the molecular history of the enzyme population.

Question 11

The graph shows oxygen uptake and blood lactate before, during and after exercise.

Which conclusion is best supported by the graph?



Programmatically generated diagram; not necessarily to scale.

- A** Only anaerobic respiration occurred during exercise because lactate increased.
- B** Only aerobic respiration occurred because oxygen uptake increased.
- C** Both aerobic and anaerobic respiration occurred, but the graph does not show that all post-exercise oxygen was used solely to remove lactate.
- D** The return of lactate to near its initial value proves that oxygen had no other post-exercise role.
- E** The graph shows that lactate was converted directly into oxygen after exercise.

WORKED SOLUTION

Rising lactate supports the occurrence of some anaerobic respiration, while oxygen uptake during exercise is consistent with aerobic respiration occurring as well. The graph does not partition post-exercise oxygen use among recovery processes, so it cannot show that all of that oxygen was used solely for lactate removal.

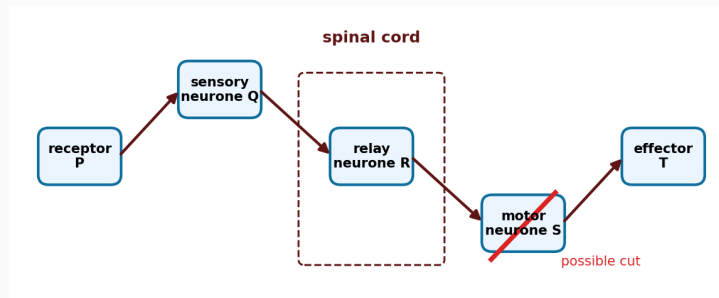
The correct option is **C**.

Distractor check: A and B treat the two pathways as mutually exclusive; D overclaims from correlation; E invents a direct conversion not represented in the graph.

Question 12

The diagram represents a withdrawal reflex. Motor neurone S is severed between the spinal cord and effector T; all other structures remain functional.

Which outcome is most likely immediately after the same stimulus is applied?



Programmatically generated diagram; not necessarily to scale.

- A** P cannot detect the stimulus, but T contracts
- B** impulses do not reach the spinal cord, but T contracts
- C** impulses reach T, but not the spinal cord
- D** sensory impulses reach the spinal cord, but T does not contract
- E** relay neurone R becomes the effector

WORKED SOLUTION

The receptor, sensory neurone and relay neurone remain intact, so the sensory signal can reach the central nervous system. The severed motor pathway cannot carry the output to the effector, so T does not contract.

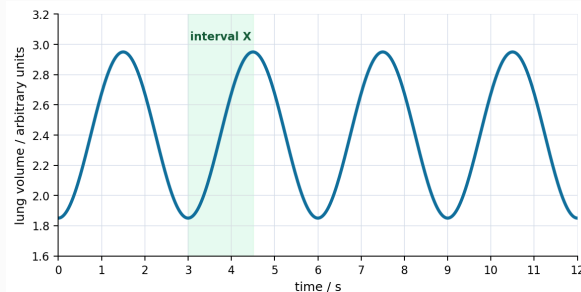
The correct option is **D**.

Distractor check: The sensory and motor limbs of the reflex must be considered separately.

Question 13

The graph shows lung volume over 12 seconds. Assume the pattern continues.

Which option correctly gives the breathing rate and describes interval X?



Programmatically generated diagram; not necessarily to scale.

- A** 5 breaths min^{-1} ; diaphragm relaxes and thoracic pressure falls
- B** 20 breaths min^{-1} ; diaphragm relaxes and thoracic pressure rises
- C** 15 breaths min^{-1} ; diaphragm contracts and thoracic pressure rises
- D** 20 breaths min^{-1} ; diaphragm contracts and thoracic pressure falls
- E** 4 breaths min^{-1} ; diaphragm relaxes and thoracic pressure is unchanged

WORKED SOLUTION

One complete cycle lasts 3 s, so the rate is $60/3=20$ breaths per minute. During X, lung volume rises: this is inspiration, when the diaphragm contracts and thoracic pressure falls below atmospheric pressure.

The correct option is **D**.

Distractor check: Counting peaks rather than complete peak-to-peak cycles can produce an incorrect rate.

Question 14

A person's heart data are shown.

condition	heart rate / beats min ⁻¹	cardiac output / dm ³ min ⁻¹
rest	75	5.25
exercise	150	15.0

The volume pumped per beat is called the *stroke volume*. Which option gives the stroke volume during exercise and the percentage increase in stroke volume from rest to exercise?

A 100 cm³; 50%

B 70 cm³; about 43%

C 100 cm³; about 43%

D 150 cm³; 100%

E 35 cm³; 30%

WORKED SOLUTION

At rest, stroke volume is $5250/75=70 \text{ cm}^3$. During exercise it is $15000/150=100 \text{ cm}^3$. The increase is

$$(100-70)/70 \times 100\% = 30/70 \times 100\% \approx 43\%.$$

The correct option is **C**.

Distractor check: A divides the 30 cm³ increase by the wrong reference value; B reports the resting stroke volume.

Question 15

A dehydrated patient releases ADH normally, but receptor molecules in the collecting ducts do not respond to ADH.

Compared with a healthy dehydrated person, which combination is most likely?

- A** low blood ADH; high water reabsorption; small volume of concentrated urine
- B** high blood ADH; high water reabsorption; small volume of concentrated urine
- C** low blood ADH; low water reabsorption; large volume of dilute urine
- D** high blood ADH; low water reabsorption; large volume of dilute urine
- E** high blood ADH; low water reabsorption; small volume of concentrated urine

WORKED SOLUTION

Dehydration lowers blood water potential, so the normal endocrine response is high ADH. If the collecting ducts cannot respond, ADH cannot produce its normal increase in water reabsorption. More water therefore remains in a larger volume of relatively dilute urine.

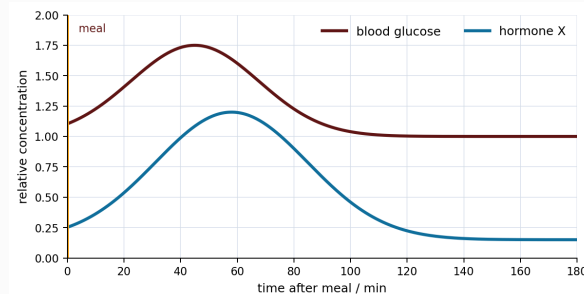
The correct option is **D**.

Distractor check: B describes a healthy response; C confuses failure of the target organ with failure to secrete the hormone.

Question 16

The graph shows blood glucose and the concentration of hormone X after a carbohydrate-rich meal. A different hormone, Y, rises during prolonged fasting and helps raise blood glucose.

Which identification and action are correct?



Programmatically generated diagram; not necessarily to scale.

- A** X is glucagon and promotes a fall in blood glucose
- B** X is insulin and promotes a fall in blood glucose
- C** X is adrenaline and Y is insulin
- D** X is insulin and Y also promotes a fall in blood glucose
- E** X is glucagon and Y is ADH

WORKED SOLUTION

Hormone X rises after blood glucose rises and acts in the negative-feedback response that brings glucose down, so X is insulin. The hormone that helps raise blood glucose during fasting is glucagon.

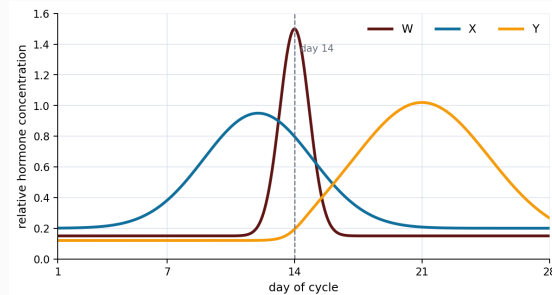
The correct option is **B**.

Distractor check: The graph shows association in time, while the stem supplies the homeostatic role needed to distinguish insulin from glucagon.

Question 17

The graph shows three hormone patterns during a menstrual cycle.

Which identification of the three hormone patterns is most consistent with the graph?



Programmatically generated diagram; not necessarily to scale.

- A** W: FSH; X: progesterone; Y: oestrogen
- B** W: progesterone; X: oestrogen; Y: LH
- C** W: LH; X: progesterone; Y: oestrogen
- D** W: oestrogen; X: LH; Y: progesterone
- E** W: LH; X: oestrogen; Y: progesterone

WORKED SOLUTION

The sharp mid-cycle surge is characteristic of LH, so W is LH. Oestrogen rises before ovulation, matching X. Progesterone is elevated mainly after ovulation and helps maintain the uterine lining, matching Y.

The correct option is **E**.

Distractor check: The peak shapes and timing, rather than the absolute vertical scale, identify the hormones.

Question 18

In a randomised trial, 18 of 600 vaccinated participants and 60 of 500 control participants developed a disease during the same follow-up period.

What is the proportional reduction in disease risk in the vaccinated group compared with the control group?

A 3%

B 9%

C 70%

D 75%

E 85%

WORKED SOLUTION

The risks are $18/600=3\%$ and $60/500=12\%$. The proportional reduction is

$$(12\%-3\%)/12\% = 3/4 = 75\%.$$

The correct option is **D**.

Distractor check: The unequal group sizes make comparing the raw case counts invalid; 9% is the absolute percentage-point reduction.

Question 19

Which statement gives the most justified interpretation of clinical-trial design?

- A** Random allocation reduces systematic group differences, double blinding reduces expectation and observer bias, and short follow-up cannot establish lifelong safety.
- B** Random allocation guarantees identical groups and therefore removes all confounding variables.
- C** Double blinding changes the biological action of the medicine and prevents adverse effects.
- D** No immediate adverse effect proves that the medicine is safe for lifelong use.
- E** Random allocation and double blinding serve the same purpose, so using both provides no additional control.

WORKED SOLUTION

Random allocation reduces systematic differences between groups but does not guarantee perfectly identical groups. Double blinding addresses expectation and observer bias. A short trial cannot establish effects that might appear only after prolonged use.

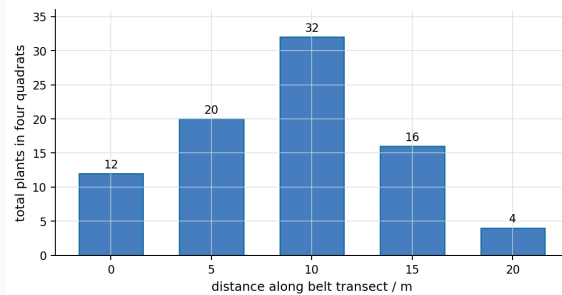
The correct option is **A**.

Distractor check: B and E overstate or merge distinct controls; C invents a physiological effect of blinding; D ignores the limit imposed by trial duration.

Question 20

At each distance on a belt transect, four quadrats of area 0.25 m^2 are sampled. The graph shows the total count in the four quadrats.

Assuming an area of 12.5 m^2 has the same mean density as the samples at 10 m, which option gives the density at 10 m and the estimated number of plants in that area?



Programmatically generated diagram; not necessarily to scale.

A 8 m^{-2} ; 100

B 32 m^{-2} ; 400

C 32 m^{-2} ; 800

D 128 m^{-2} ; 1600

E 8 m^{-2} ; 400

WORKED SOLUTION

The four quadrats together sample $4(0.25)=1.00 \text{ m}^2$. A count of 32 therefore gives a density of 32 m^{-2} . The estimated population is $32(12.5)=400$.

The correct option is **B**.

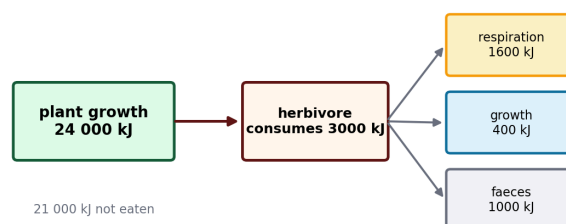
Distractor check: A divides the count by four but forgets that each quadrat is only one-quarter of a square metre; D multiplies instead of combining areas.

Question 21

The diagram shows annual energy flow from 24 000 kJ of plant growth. A herbivore consumes 3000 kJ; 1000 kJ leaves in faeces, 1600 kJ is transferred by respiration and 400 kJ becomes new herbivore biomass. For this question, **absorbed energy** means consumed energy minus energy lost in faeces.

1. The energy absorbed by the herbivore is 2000 kJ.
2. Twenty per cent of the absorbed energy becomes new herbivore biomass.
3. If all uneaten plant material and faeces enter decomposer food chains, they receive 11/12 of the plant-growth energy shown.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

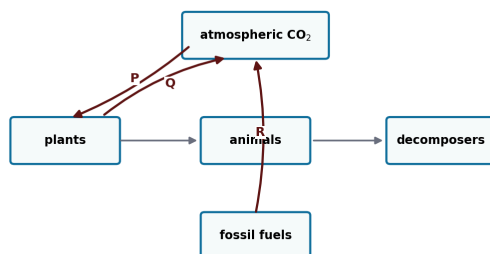
Using the definition in the stem, absorbed energy is $3000 - 1000 = 2000$ kJ. Growth efficiency relative to absorbed energy is $400/2000 = 20\%$. Uneaten plant material is $24000 - 3000 = 21000$ kJ; adding faeces gives 22 000 kJ, and $22000/24000 = 11/12$.

All three statements are correct, so the answer is **H**.

Distractor check: The denominator changes between consumption and absorbed-energy efficiencies; the energy-flow diagram must be read before calculating.

Question 22

The diagram shows part of the carbon cycle. Which row correctly identifies processes P, Q and R?



Programmatically generated diagram; not necessarily to scale.

A P photosynthesis; Q respiration; R combustion

B P respiration; Q photosynthesis; R decomposition

C P transpiration; Q respiration; R combustion

D P photosynthesis; Q translocation; R respiration

E P combustion; Q photosynthesis; R decomposition

WORKED SOLUTION

P transfers carbon dioxide from the atmosphere into plant biomass by photosynthesis. Q returns carbon from organisms to atmospheric carbon dioxide by respiration. R transfers carbon from fossil fuels to atmospheric carbon dioxide by combustion.

The correct option is **A**.

Distractor check: The arrow direction is decisive: respiration and photosynthesis move carbon in opposite directions.

Question 23

Three identical lake microcosms are maintained for two weeks. Assume that the decomposer inhibitor does not directly affect algal growth, light transmission or oxygen solubility.

treatment	algal biomass / units	light at submerged plants / %	dissolved oxygen / mg dm ⁻³
control	2	80	8
nitrate added	10	25	3
nitrate + decomposer inhibitor	9	28	7

1. The low light in the nitrate treatment is mainly caused by decomposer activity.
2. Under these conditions, added nitrate is associated with increased algal biomass.
3. Restoring much of the oxygen without restoring much light supports the conclusion that decomposition contributes to oxygen loss.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Blocking decomposers barely changes algal biomass or light, so statement 1 is false: decomposers are not the main explanation for shading. Nitrate addition is followed by a fivefold algal increase, so statement 2 is supported. The inhibitor raises oxygen from 3 to 7 mg dm⁻³ while light remains low, supporting statement 3 and a role for decomposer respiration in oxygen loss.

Statements 2 and 3 only are correct, so the answer is **G**.

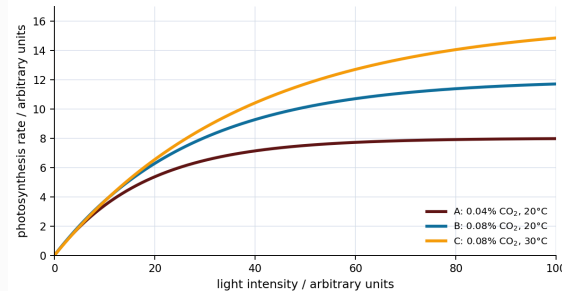
Distractor check: The controlled comparison is nitrate versus nitrate plus inhibitor, not nitrate versus control alone.

Question 24

The graph shows photosynthesis rate under three combinations of carbon-dioxide concentration and temperature.

1. At low light intensity, light is a limiting factor in all three conditions.
2. At high light intensity and 20°C, the comparison of A with B shows carbon dioxide limiting the rate in A.
3. Raising the temperature from 20°C to 30°C produces the same absolute increase in photosynthesis rate at every light intensity shown.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

All curves initially rise with light, showing light limitation. At high light, B exceeds A when only carbon dioxide concentration has changed, so carbon dioxide limits A there. Statement 3 is false: the vertical separation between B and C varies with light intensity and is zero at zero light, so the temperature increase does not produce the same absolute rate increase throughout.

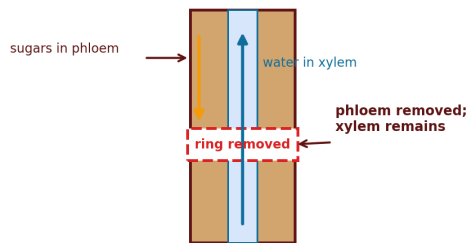
Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: The size of a temperature effect must be read from the vertical separation of B and C; it is not constant across the graph.

Question 25

A narrow ring of tissue containing phloem is removed from a woody stem. The xylem remains intact.

Which outcome is most likely?



Programmatically generated diagram; not necessarily to scale.

- A** Upward water movement stops immediately, and sugars accumulate below the ring.
- B** Upward water movement may continue initially, sugars accumulate above the ring, and root respiration may later decrease.
- C** Phloem continues carrying sugars to the roots because the xylem is intact.
- D** Sugars accumulate below the ring, causing an immediate increase in root growth.
- E** Neither transport system is affected because both xylem and phloem lie deeper than the removed ring.

WORKED SOLUTION

The intact xylem can continue transporting water and mineral ions upward immediately after girdling. Removing phloem interrupts translocation from the leaves toward the roots, so sugars accumulate above the ring and the roots later receive less respiratory substrate.

The correct option is **B**.

Distractor check: Xylem and phloem directions/functions must not be interchanged, and the word 'immediately' matters.

Question 26

A circular plasmid has one recognition site for restriction enzyme E. A linear donor DNA molecule carries gene G and has an E recognition site at each end. Both DNA molecules are completely cut with E and then mixed with DNA ligase. Assume that every E cut produces compatible sticky ends.

1. Cutting the plasmid once converts it into one linear DNA molecule.
2. A donor fragment carrying G can become joined into the plasmid.
3. The donor fragment can join into the plasmid in only one orientation.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A single cut opens a circular plasmid to give one linear DNA molecule, so statement 1 is correct. Compatible sticky ends allow the donor fragment to ligate into the opened plasmid, so statement 2 is correct. Because the same compatible end is present at each end of the donor, it can be inserted in either orientation; statement 3 is false.

Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: Compatible sticky ends permit joining but do not by themselves impose a direction on the insert.

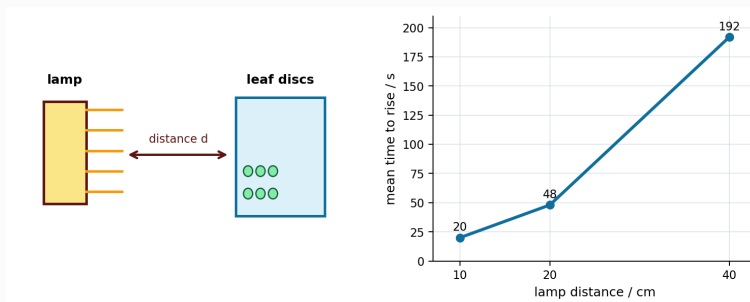
Question 27

Identical leaf discs are placed in identical solutions at constant temperature and carbon-dioxide concentration. Mean time to rise is used as an inverse measure of photosynthesis rate. Light intensity from the lamp is proportional to $1/d^2$, where d is lamp distance.

The observed mean times are 20 s at 10 cm, 48 s at 20 cm and 192 s at 40 cm.

1. If light alone limits the rate between 20 cm and 40 cm, the predicted mean time at 30 cm is 108 s.
2. The inverse-square prediction from 20 cm to 10 cm is 12 s, so the observed 20 s is consistent with another factor becoming limiting at high light intensity.
3. Raising the temperature would reduce the 10 cm mean time to the inverse-square prediction of 12 s.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Because rate is proportional to $1/\text{time}$ and light to $1/d^2$, time is proportional to d^2 while light is the only limiting factor. Thus $48(30/20)^2 = 108$ s. Moving from 20 cm to 10 cm predicts $48(10/20)^2 = 12$ s, but the observed time is longer; another factor could be limiting. The data do not identify temperature as that factor or determine the result of changing it.

Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: The calculation tests inverse proportionality; the interpretation distinguishes a possible explanation from one established by the data.